

ChE 201 – Introduction to Thermodynamics –Spring 2026

Student Learning Outcome 1

Rubric for assessment problem on Midterm 2 (Problem #2)

Application of energy balance to a piston/cylinder device with air.

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem 2a, 2b 4 pts	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (<2 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (2 pts)	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (3-4 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem 2e, 2f, 25pts	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (<10 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (10-14 pts)	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (15-25 pts)
4. Translates academic theory into engineering applications Problem 2c, 2d 6 pts	Does not appear to grasp the connection between theory and the problem (< 3 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (3 pts)	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (4-6 pts)

Averages:

Performance Indicators: (1) 2 (2) 2.50 (4) 2.42 Overall Average: 2.64

Average scores were fairly uniform although there was wide variation among students. Clearly, most students were strong in formulating the engineering system and model equations, and a majority 10/12 were satisfactory or higher in applying the engineering principles to solve the three problems.

CHE 201 MidTerm Exam 2

March 11th, 2026

QUESTION 2: (35 pts)

A piston cylinder device initially contains 3m^3 of air at 500kPa and 150°C . The air is now expanded to 10m^3 , during which the pressure-volume relationship is $PV^{1.5} = \text{constant}$. The air is modeled as an ideal gas and kinetic and potential energy effects are negligible.

- (2 pts) Draw a schematic of the system and label the system boundary.
- (2 pts) State the engineering model
- (2 pts) Write the simplified energy balance and compressibility factor (Z).
- (4pts) Draw P-v diagram (Pressure-specific volume)
- (10 pts) Determine the work (kJ)
- (15 pts) Determine the heat (kJ) (i) by using Table, (ii) by using constant specific heat values, (iii) by using $k = 1.4$

	c_p	c_v	k
Temp. K	Air		
250	1.003	0.716	1.401
300	1.005	0.718	1.400
350	1.008	0.721	1.398
400	1.013	0.726	1.395
450	1.020	0.733	1.391
500	1.029	0.742	1.387
550	1.040	0.753	1.381
600	1.051	0.764	1.376
650	1.063	0.776	1.370
700	1.075	0.788	1.364
750	1.087	0.800	1.359

Solution

b) Engineering Model

$$PV^{1.5} = \text{constant}$$

$$KE = 0, PE = 0$$

Ideal gas

closed system

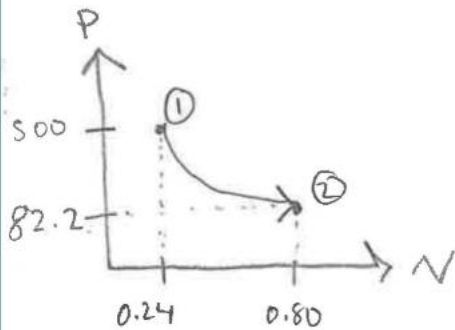
$$c) \Delta E = Q - W$$

$$\Delta E = \Delta u + \cancel{\Delta KE} + \cancel{\Delta PE}$$

$$\Delta u = Q - W$$

$Z = 1$ for
ideal gas

d) P-V



$$P_1 V_1^{1.5} = P_2 V_2^{1.5}$$

$$500 \text{ kPa} \times (3 \text{ m}^3)^{1.5} = P_2 (10 \text{ m}^3)^{1.5}$$

$$P_2 = 82.2 \text{ kPa}$$

$$P V = n R T \rightarrow n = \frac{8.314 \text{ kJ/kmol} \cdot \text{K} \times (150 + 273) \text{ K}}{28.97 \text{ kg/kmol} \times 500 \text{ kPa}} = 0.24 \frac{\text{m}^3}{\text{kg}}$$

$$n_1 = 0.24 \frac{\text{m}^3}{\text{kg}}$$

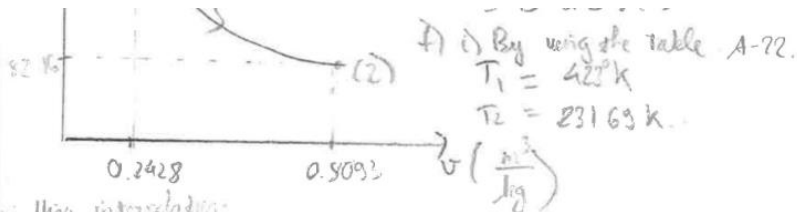
$$500 \text{ kPa} \times (0.24 \frac{\text{m}^3}{\text{kg}})^{1.5} = 82.2 \text{ kPa} (V_2)^{1.5}$$

$$V_2 = 0.80 \frac{\text{m}^3}{\text{kg}}$$

$$e) \quad W = \int_3^{10} P dv = \frac{P_2 V_2 - P_1 V_1}{-n+1}$$

$$W = \frac{(82.2 \text{ kPa} \times 10 \text{ m}^3) - (500 \text{ kPa} \times 3 \text{ m}^3)}{-1.5 + 1}$$

$$W = 1356 \text{ kJ}$$



A) By using the table A-22.
 $T_1 = 423 \text{ K}$
 $T_2 = 231.69 \text{ K}$

Using interpolation:

$$\frac{430 - 420}{307.99 - 302.96} = \frac{430 - 423}{307.99 - u_{T1}} \Rightarrow u_{T1} = 302.88 \left(\frac{kJ}{kg} \right)$$

$$\frac{240 - 230}{171.13 - 164} = \frac{240 - 231.69}{171.13 - u_{T2}} \Rightarrow u_{T2} = 167.70 \left(\frac{kJ}{kg} \right)$$

$$\Rightarrow \Delta u = \Delta u \cdot m = (u_2 - u_1) m = (167.70 - 302.88) \times 12.36 = -1670.85 \text{ kJ}$$

$$\Delta u = Q - W \Rightarrow Q = \Delta u + W = -1670.85 \text{ kJ} + 1356.84 \text{ kJ} = -314.05 \text{ kJ}$$

(ii) By using specific heat:

$$\text{At } T_1 = 423 \text{ K} \Rightarrow C_v = 0.73 \frac{kJ}{kg \cdot K} \Rightarrow C_v = 0.722 \frac{kJ}{kg \cdot K}$$

$$T_2 = 231.69 \text{ K} \Rightarrow C_v = 0.714 \frac{kJ}{kg \cdot K}$$

At constant C_v

$$\Rightarrow C_v = \frac{8}{5} R \Rightarrow \Delta u = C_v \Delta T = 0.722 \frac{kJ}{kg \cdot K} (231.69 \text{ K} - 423 \text{ K})$$

$$= -138.13 \frac{kJ}{kg}$$

$$\Rightarrow \Delta u = \Delta u \cdot m = -138.13 \times 12.36 = -1707.66 \text{ kJ}$$

$$\Rightarrow Q = \Delta u + W = -1707.66 + 1356.8 = -350.86 \text{ kJ}$$

(iii) By $k = 1.4 = \frac{C_p}{C_v}$

$$C_p = C_v + R \Rightarrow \frac{C_p}{C_v} = \frac{C_v}{C_v} + \frac{R}{C_v} \Rightarrow 1.4 = 1 + \frac{R}{C_v}$$

$$\Rightarrow C_v = 0.717 \frac{kJ}{kg \cdot K}$$

$$\Rightarrow \Delta u = C_v \Delta T = 0.717 \frac{kJ}{kg \cdot K} (231.69 \text{ K} - 423 \text{ K}) = -137.26 \frac{kJ}{kg}$$

$$\Rightarrow \Delta u = -1696.5 \text{ kJ}$$

$$\Rightarrow Q = \Delta u + W = -339.7 \text{ kJ}$$